# Module 1 — Read 2

**Binomial test workflow and R**

# Module 1, Read 2: The binomial test in practice

**Project:** Project 1 — Testing a Claim · **Prerequisite:** Read 1 and Quiz A · **Next:** Quiz B, demo lab, exercise lab.

This reading is your **self-paced textbook** for conducting Project 1. Complete Read 1 and Quiz A first. Use **Check yourself** boxes before Quiz B.

## Learning outcomes

Three sections walk you from hypotheses through the full binomial test workflow in R.

After this reading you should be able to:

### Section 1: Hypotheses and the binomial test

*Introduction to the binomial test (ESP example)*

- Translate a research question about a proportion into $H_0$ and $H_a$.
- Check the **four BRP conditions** for data from a fixed number of binary trials.
- Connect the ESP activity to the structure of **Project 1**.

### Section 2: ESP worksheet workflow

*ESP worksheet — binomial test in practice*

- State hypotheses and verify the BRP conditions for the ESP class example.
- Identify $n$ and $p$ under $H_0$ and record the observed count $x$.
- Obtain a **p-value** in R using the same workflow as Project 1.

### Section 3: Ten-step workflow and worked examples

*Binomial test — steps and worked examples*

- Carry out the full workflow: question, parameter, hypotheses, $n$ and $p$ under $H_0$, BRP check, null distribution plot, observed $x$, shading by $H_a$, p-value, and conclusion in context.
- Handle **one-sided** and **two-sided** alternatives correctly when shading and interpreting the p-value.

> ⚠ Do not use ESP as your Project 1 topic
>
> The ESP examples below are **class models only**. Choose your own binomial process ($n = 25$) that is not a lecture example.

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## Section 1: Hypotheses and the binomial test

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# A motivating question: Do I have ESP?

**Extra Sensory Perception (ESP)** is the ability to accurately predict what will happen using supernatural methods. Do I have it? If so, I could make lots of money by betting on the stock market! To find out, we can take an online ESP test and then analyze the data using the **Binomial Test**.

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# The online ESP test

We will collect data from the **Advanced ESP (Zener Cards) test** at PsychicScience.org.

- The test uses **5 possible symbols** (circle, cross, waves, square, star).
- On each trial, the computer randomly selects one symbol; your task is to guess it.
- We will use **25 attempts** (25 cards).
- Choose: **Clairvoyance**, **Open deck**, **Cards seen**, and **25 cards**.

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# From question to hypotheses

> **!** Notation: Parameter $p$ and hypotheses
>
> Let $p$ be the probability of a correct prediction.
>
> - If I **don't** have ESP, I'm just guessing among 5 options, so $p = 1/5 = 0.2$.
> - If I **do** have ESP, I should get more than chance, so $p > 0.2$.
>
> In conventional notation:
>
> $$H_0 : p = 0.2 \quad \text{vs.} \quad H_a : p > 0.2$$
>
> - **Null hypothesis** ($H_0$): baseline model with an **equality** (here, $p = 0.2$).
> - **Alternative hypothesis** ($H_a$): research question as an **inequality** (here, $p > 0.2$).

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# Data collection and the four conditions

When we record the number of correct predictions out of 25 attempts, we treat this as a **Binomial Random Process**. We check the 4 conditions:

1. **Binary outcome:** Each prediction was either correct or not.
2. **Fixed number of trials:** We made a fixed number of attempts, $n = 25$.
3. **Fixed probability:** The probability of a correct guess on each trial doesn't change (open deck: each card is chosen at random from 5 options each time).
4. **Independent trials:** We don't use information from previous attempts to inform the next guess (and the computer randomizes each card independently).

So the number of correct predictions in 25 attempts can be modeled by a **binomial random variable** with $n = 25$ and $p = 0.2$ under the null hypothesis.

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# Sample project structure (Project 1)

The following is an example of how you might write up a **Binomial Test** in your one-page Project 1 summary. This example uses the ESP question and data from the online test.

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## Introduction

Extra Sensory Perception (ESP) is the ability to accurately predict what will happen using supernatural methods. Do I have it? If so, I could make lots of money by betting on the stock market! To find out, I took an online ESP test and then analyzed the data using the Binomial Test.

Let $p$ be the probability that I have an accurate prediction. In the online ESP test, there were 5 possible predictions, so if I don't have ESP, then $p = 1/5$. If I do have ESP, then $p > 1/5$. These hypotheses in conventional notation are:

$$H_0 : p = 0.2 \quad \text{vs.} \quad H_a : p > 0.2$$

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## Data Collection

I recorded the number of correct predictions out of 25 attempts in the online ESP test at Advanced ESP card guessing (Zener Cards) test — PsychicScience.org.

I believe this data collection satisfied the 4 conditions of a Binomial Random Process:

1. Each prediction was correct or not (**binary outcome**).
2. I made a fixed number of attempts, $n = 25$ (**fixed number of trials**).
3. My probability of getting a prediction correct shouldn't change between attempts (**fixed probability of success** in a single trial).
4. I didn't use information from my previous attempts to inform my predictions (**independent trials**).

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## Data Analysis

**Descriptive statistics:** In 25 attempts, I correctly predicted the shape on 4 cards (16%).

**Inferential statistics:** Let $X$ be the number of correct predictions in 25 attempts. Assuming I'm just guessing, $X$ is a Binomial random variable with $n = 25$ and $p = 0.2$. I computed the probability of each value of $X$ and the probability of seeing my data (4) **or more** correct predictions using R. This **p-value** is 0.76.

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## Conclusion

There is a 76% chance of seeing my results of 4 correct predictions or more in another set of 25 attempts if I don't have ESP and am just guessing. Since this is a large probability, I've concluded I don't have ESP and will find other ways besides betting to make money.

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## Takeaway

> **!** Definition: p-value
>
> Under the **null model** ($H_0$), the **p-value** is the probability of observing data **as or more extreme than** what you observed (in the direction of $H_a$), **assuming $H_0$ is true**.
>
> - A **small** p-value suggests the null model is not consistent with the data.
> - A **large** p-value (as in this example) suggests the data are consistent with just guessing.

- Your Project 1 report should follow this same structure: Introduction (question, $p$, hypotheses), Data Collection (how you collected data and why the 4 BRP conditions hold), Data Analysis (descriptive and inferential statistics, including p-value), and Conclusion.

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> **!** Seven steps of the binomial test
>
> 1. **Define the parameter of interest in your research question,** $p$: Identify and define $p$. Here $p$ is the **theoretical** probability of a "success" in a single trial for the population or process (e.g., the probability of a correct guess on one ESP card). $p$ is a parameter of the process, not a value observed in your sample.
> 2. **State the null and alternative hypotheses using appropriate symbols:**
>    - **Alternative hypothesis ($H_a$)**: The **direction** (inequality) of your best guess for the range of values of $p$ — your research question in mathematical form (e.g., $p > 0.2$, $p < 0.2$, or $p \neq 0.2$).
>    - **Null hypothesis ($H_0$)**: Take the alternative and change the inequality to an **equality** (e.g., if $H_a : p > 0.2$, then $H_0 : p = 0.2$). This is the baseline or "no effect" assumption.
> 3. **Assuming the null hypothesis is true, generate the null distribution of** $x$ for sample size $n$: Build the probability distribution of the number of successes, $x$, you would expect to see in $n$ trials if $H_0$ were true. This null distribution is used to compute the p-value.
> 4. **Verify that the 4 conditions of a Binomial Random Process (BRP) are satisfied by the data collection:** Confirm that (a) each trial has a **binary** outcome (success or failure), (b) the number of trials $n$ is **fixed**, (c) the probability of success $p$ is **fixed** on every trial, and (d) trials are **independent**.
> 5. **Identify the value of** $x$ from the dataset: Record the **observed** number of successes in your $n$ trials (e.g., number of correct ESP guesses out of 25).
> 6. **Determine whether the data are consistent with the null hypothesis by computing the p-value:** The **p-value** is the probability of observing **new** data (in $n$ trials) that is equal to or **more extreme** (in the direction of the alternative) than the data you actually observed, **assuming the null hypothesis is true**. A small p-value suggests the observed data are inconsistent with the null; a large p-value suggests they are consistent.
> 7. **Make a conclusion in context:** Interpret the p-value and state a conclusion about your research question (e.g., whether there is evidence for ESP or whether the results are consistent with chance).

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## R code: Colab notebook

You can practice binomial test code in the Module 1 demo lab and R reference. A sample Colab notebook is also available:

**Sample binomial test notebook (Colab)** ::: {.callout-tip icon=false} ## Check yourself — Hypotheses and the binomial test

1. ESP test: 25 trials, 5 symbols, guessing under $H_0$. Write $H_0$ and $H_a$ for testing ESP.
2. In words, what does $p$ represent in Project 1?

3. State the definition of **p-value** for a binomial test.
4. Why does the ESP data collection satisfy the four BRP conditions?

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**Answers — Hypotheses and the binomial test**

1. $H_0 : p = 0.2$ vs $H_a : p > 0.2$ where $p$ = probability of a correct guess on one trial.
2. The **theoretical** probability of success on **one trial** for the population/process — not the observed $\hat{p}$ from your sample.
3. Probability of data **as or more extreme than observed** (in direction of $H_a$), **assuming $H_0$ is true**.
4. Binary (correct/incorrect); fixed $n = 25$; fixed $p = 0.2$ under null (open deck); independent guesses.

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## Section 2: ESP worksheet workflow

Follow the steps below to walk through the ESP class example, compute a p-value in R, and see how the pieces fit Project 1. This follows the same structure as Section 1 and the sample report (PDF).

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## Step 1: State your null and alternative hypotheses (ESP)

You are testing whether you have ESP using the same setup as the instructor ESP example in Section 1.

**!** Notation: Parameter $p$

- Let $p$ = the probability that **you** correctly predict the symbol on a single trial.
- If you don't have ESP, you're guessing among 5 symbols, so $p = 0.2$. If you do have ESP, $p > 0.2$.

**Your turn.** Write the null and alternative hypotheses and double-check that they match the hypotheses in Section 1:

$H_0$: $p = 0.2$

$H_a$: $p > 0.2$

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## Step 2: Verify the 4 conditions of a Binomial Random Process (in context)

Before we treat the number of correct guesses as a binomial random variable, we need to check that the 4 BRP conditions hold for this data collection.

**!** Checklist: Four BRP conditions (in context)

**Your turn.** In the context of the online ESP test (25 attempts, 5 symbols, open deck, cards seen), briefly think through why each condition is satisfied:

1. **Binary outcome:** Each trial has only two outcomes. Why? (What are they?)
2. **Fixed number of trials:** What is $n$? How do we know it's fixed?
3. **Fixed probability of success:** Under the null hypothesis that you are guessing, what is the value of $p$ on each trial? Why doesn't it change from trial to trial in the "open deck" setup?
4. **Independent trials:** Why doesn't one guess affect the next? Of if you believe guesses are not independent, what could you do to collect data in a way that your guess on one attempt doesn't affect your guess on another attempt?

**Model answers — BRP checklist (ESP)**

1. **Binary:** each guess is correct or incorrect.

2. **Fixed $n$:** you commit to 25 attempts before starting.
3. **Fixed $p$:** under $H_0$, each guess has probability 0.2 (open deck re-randomizes each trial).
4. **Independent:** guesses should not use prior cards; if worried, take breaks or blind yourself to prior results.

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## Step 3: Identify $n$ and $p$ (under the null)

**Your turn.** For this activity we will all use 25 attempts. If the null hypothesis is true that **you** don't have ESP then

- $n = \mathbf{25}$
- $p = \mathbf{0.2}$

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## Step 4: Collect your data — take the ESP test

1. Go to the **Advanced ESP (Zener Cards) test**: https://psychicscience.org/esp3.
2. Watch the ad to access the test for "free",
3. Use the default options: **Clairvoyance**, **Open deck**, **Cards seen**, **25 cards**.
4. Complete all 25 guesses.
5. Record your **number of correct predictions** (this is your observed value of $x$).

**Your** $x$ (number of correct predictions out of 25): *(record your own result after taking the test)*

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## Step 5: Find your p-value using the sample R notebook

1. Open the Module 1 demo lab or sample Colab notebook: **Sample binomial test notebook (Colab)**.
2. In the notebook, locate the observed number of correct predictions (the instructor example used $x = 4$).
3. **Change all the values of $x$ from 4 to your value of $x$** (from Step 4).
4. **Run all the code** in the notebook (e.g., Runtime → Run all, or run each cell in order).
5. Find the **p-value** reported by the notebook and record it.

**Your p-value:** *(from the Colab notebook after you enter your $x$)*

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## Step 6: Post your results on Canvas

Go to the **Canvas discussion thread** for this activity.

Post the following:

- Your $x$ (number of correct predictions out of 25).
- Your **p-value**. We'll see who in class is **least consistent** with the hypothesis of no ESP by seeing who has the **smallest p-value** (or equivalently, the **largest** $x$). This will be the person who is most likely to have ESP! ::: {.callout-tip icon=false} ## Check yourself — ESP worksheet workflow

1. Under the null (no ESP), what are $n$ and $p$ for the 25-trial Zener test?
2. What is your observed $x$ after the test? (Use your own data.)
3. Name the four BRP conditions you must defend in your Project 1 report.

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**Answers — ESP worksheet workflow**

1. $n = 25$; $p = 0.2$.
2. Answers vary; $x$ = your number of correct guesses out of 25.
3. Binary outcome; fixed $n$; fixed $p$ on each trial; independent trials.

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## Section 3: Ten-step workflow and worked examples

**!** Ten steps of the binomial test (detailed workflow)

For each problem, follow these steps:

1. State the research question.
2. Define the parameter of interest, $p$, in words.
3. State the null and alternative hypotheses using appropriate symbols.
4. Identify $n$ and the value of $p$ under $H_0$ (for the null distribution).
5. Verify that the 4 conditions of a Binomial Random Process (BRP) are satisfied by the data collection.
6. Assuming the null hypothesis is true, create a plot of $x$ vs $\text{dbinom}(x, n, p)$ (the null distribution) in R — same pattern as Project 1 (`geom_col` + `dbinom` heights).
7. Identify the observed value of $x$ from the dataset.
8. On the plot, add `geom_vline` at the observed $x$ and shade the bar(s) on the **same side(s) as the inequality in** $H_a$ (coral bars in the examples below).
9. Compute the **p-value**: the probability of observing data as or more extreme than the observed $x$ in the direction of $H_a$, assuming $H_0$ is true. (For a two-sided $H_a$, use both tails.)
10. Make a conclusion in context (reject $H_0$ or fail to reject $H_0$, and what that means for the research question).

**!** Definition: p-value (binomial test)

The **p-value** is the probability of observing data **as or more extreme than** the observed $x$ **in the direction of** $H_a$**, assuming** $H_0$ **is true**. For a two-sided $H_a$, sum probabilities in **both tails**.

**!** How to read the shaded null plots

Each figure below is the **null distribution** ($x$ vs $P(X = x)$ under $H_0$).

1. **Vertical dashed line** — your **observed** count $x$ from the data (the cut at the data you collected).
2. **Coral (red) bars** — the **p-value region**: bar heights you **add** for counts at least as extreme as yours **on the same side(s) as the inequality in** $H_a$:
   - $H_a : p > p_0 \rightarrow$ shade the **right** tail ($x \geq$ observed).
   - $H_a : p < p_0 \rightarrow$ shade the **left** tail ($x \leq$ observed).
   - $H_a : p \neq p_0 \rightarrow$ shade **both** tails (low counts and high counts that are as or more extreme).
3. **Steelblue bars** — counts that are **not** in the p-value region (still part of the null model, but not "as extreme" in the direction of $H_a$).

The **p-value** equals the sum of $P(X = x)$ over the shaded bars. Use the same R pattern in Project 1 and the demo lab. Figures render when this page is built; the R code is hidden below each plot.

**!** Notation in worked examples

- **Parameter** $p$: theoretical probability of success on one trial (define in words for your question).
- **Observed data** $x$: number of successes in $n$ trials.

- **Hypotheses:** $H_0$ uses an **equality**; $H_a$ uses $<$, $>$, or $\neq$.

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## Example 1:

**Research question:** Is the proportion of left-handed students at our school higher than the national rate of 10%?

**Parameter:** $p$ = probability that a randomly chosen student from our school is left-handed (i.e., the proportion of left-handers in the population we are sampling).

**Hypotheses:** $H_0 : p = 0.10$ vs $H_a : p > 0.10$

$n$ and $p$ under $H_0$: $n = 25$ students, $p = 0.10$

**BRP conditions (brief):** Binary (left-handed or not), fixed $n = 25$, fixed $p$ under null, independent students.

**Null plot:** $H_a : p > 0.10$ — shade where $x \geq 6$ (same side as $>$ in the alternative). Dashed line at observed $x = 6$.

### Code anatomy — Null plot with p-value shading (Project 1 pattern)

| Part | Role |
|---|---|
| n `25` | Number of trials under $H_0$ |
| p `0.10` | Success probability under $H_0$ |
| x_obs `6` | Observed count from **your data** |
| in_pvalue | Logical vector: `TRUE` where bar is in p-value region ($x \geq$ observed (right tail)) |
| geom_col(aes(fill = in_pvalue)) | Bar heights from `dbinom`; color marks p-value region |
| geom_vline(xintercept = x_obs) | Vertical line at **observed** $x$ |
| Sum of coral bar heights | Approximates the p-value (check with `pbinom`) |

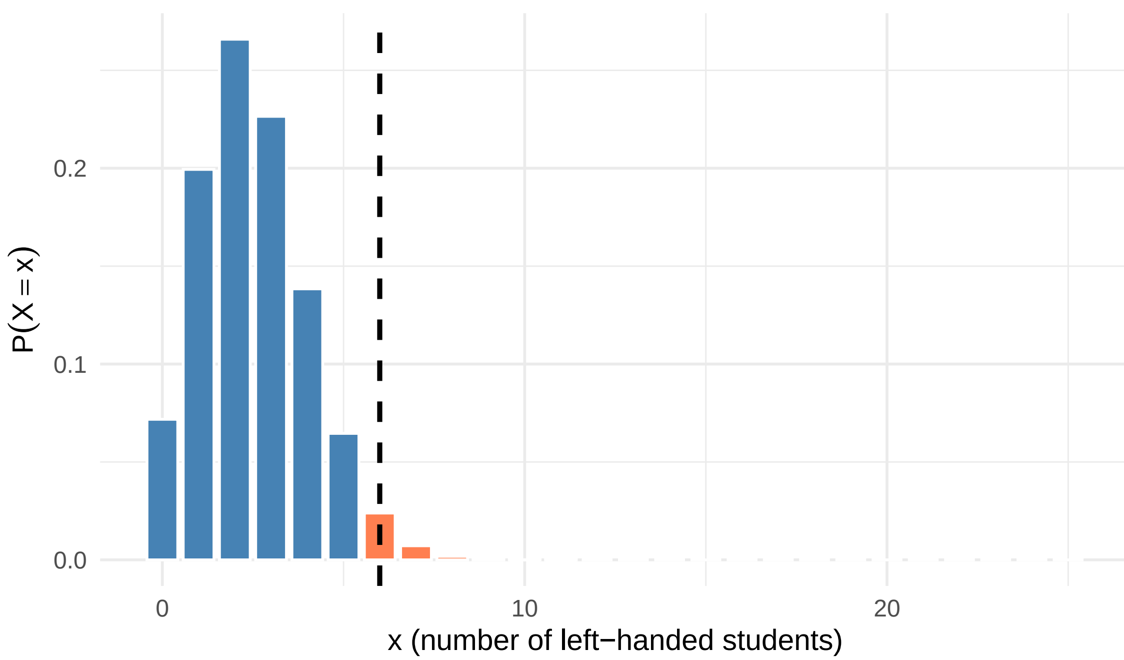


Figure 1: Example 1: left-handed students — p-value region shaded for Ha: p > 0.10, x = 6

**Data:** In a sample of 25 students, $x = 6$ were left-handed.

**p-value:** $P(X \geq 6 \mid H_0) = 1 - P(X \leq 5)$. For example, `1 - pbinom(5, 25, 0.10)` $\approx 0.03$. Small p-value.

**Conclusion:** Reject $H_0$. The data are inconsistent with a 10% left-handed rate; we have evidence that the proportion of left-handed students at our school is higher than 10%.

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# Example 2

**Research question:** Is the proportion of defective items from the new supplier less than 15%?

**Parameter:** $p$ = probability that a randomly selected item from the new supplier is defective (i.e., the proportion of defectives in the process).

**Hypotheses:** $H_0 : p = 0.15$ vs $H_a : p < 0.15$

$n$ and $p$ under $H_0$: $n = 25$ items, $p = 0.15$

**BRP conditions (brief):** Binary (defective or not), fixed $n = 25$, fixed $p$ under null, independent items.

**Data:** In a sample of 25 items, $x = 3$ were defective.

**Null plot:** $H_a : p < 0.15$ — shade where $x \leq 3$ (same side as < in the alternative). Dashed line at observed $x = 3$.

## Code anatomy — Null plot with p-value shading (Project 1 pattern)

| Part | Role |
|---|---|
| n `25` | Number of trials under $H_0$ |
| p `0.15` | Success probability under $H_0$ |
| x_obs `3` | Observed count from **your data** |
| in_pvalue | Logical vector: `TRUE` where bar is in p-value region ($x \leq$ observed (left tail)) |
| geom_col(aes(fill = in_pvalue)) | Bar heights from `dbinom`; color marks p-value region |
| geom_vline(xintercept = x_obs) | Vertical line at **observed** $x$ |
| Sum of coral bar heights | Approximates the p-value (check with `pbinom`) |

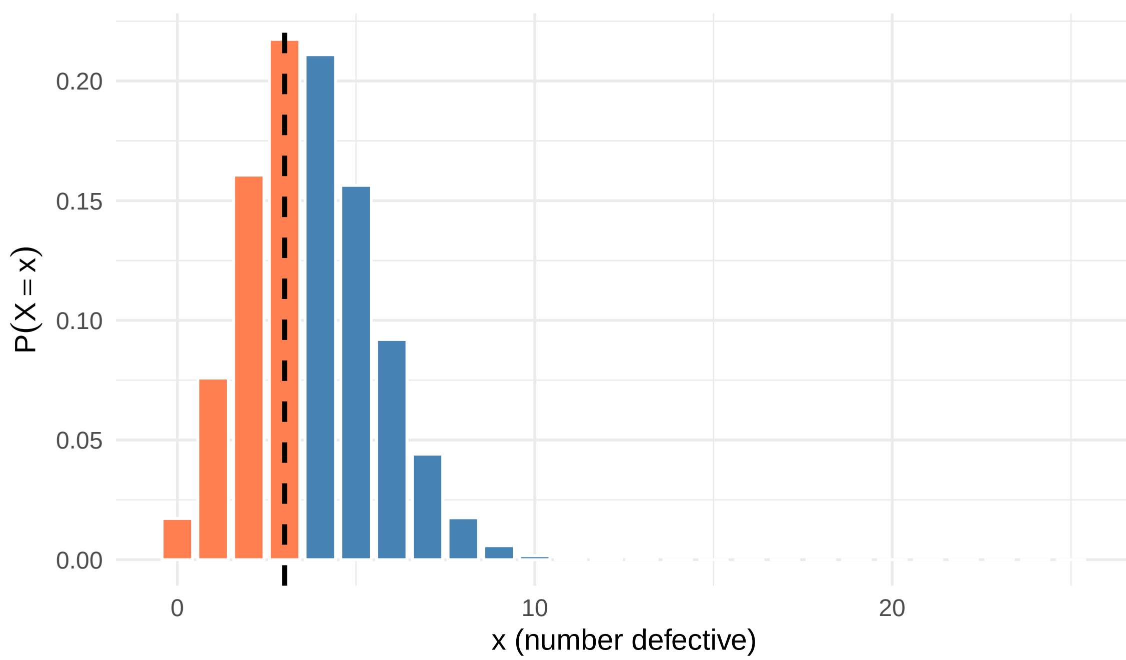


Figure 2: Example 2: defective items — p-value region shaded for Ha: p < 0.15, x = 3

**p-value:** $P(X \leq 3 \mid H_0) = \text{pbinom}(3, 25, 0.15) \approx 0.47$. Large p-value.

**Conclusion:** Fail to reject $H_0$. The data are consistent with a 15% defective rate; we do not have evidence that the proportion of defectives from the new supplier is less than 15%.

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# Example 3

**Research question:** Are the two dice fair, in terms of the probability that the sum is 7 or 11? (With fair dice, that probability is 8/36: 6 ways to sum to 7 plus 2 ways to sum to 11, out of 36 equally likely outcomes.)

**Parameter:** $p$ = probability that a single roll of two dice gives a sum of 7 or 11.

**Hypotheses:** $H_0 : p = 8/36$ vs $H_a : p \neq 8/36$

$n$ and $p$ under $H_0$: $n = 25$ rolls of the two dice, $p = 8/36$ (under the null that the dice are fair).

**BRP conditions (brief):** Binary (sum is 7 or 11, or not), fixed $n = 25$, fixed $p$ under null, independent rolls.

**Data:** In 25 rolls of the two dice, $x = 2$ times the sum was 7 or 11 (far from the expected $25 \times (8/36) \approx 5.6$ under $H_0$).

**Null plot:** $H_a : p \neq 8/36$ — shade **both** tails ($x \leq 2$ and $x \geq 9$ for this observed $x = 2$). Dashed line at observed $x = 2$.

## Code anatomy — Null plot with p-value shading (Project 1 pattern)

| Part | Role |
|---|---|
| n `25` | Number of trials under $H_0$ |
| p `8/36` | Success probability under $H_0$ |
| x_obs `2` | Observed count from **your data** |
| in_pvalue | Logical vector: `TRUE` where bar is in p-value region (both tails for two-sided $H_a$) |
| geom_col(aes(fill = in_pvalue)) | Bar heights from `dbinom`; color marks p-value region |
| geom_vline(xintercept = x_obs) | Vertical line at **observed** $x$ |
| Sum of coral bar heights | Approximates the p-value (check with `pbinom`) |

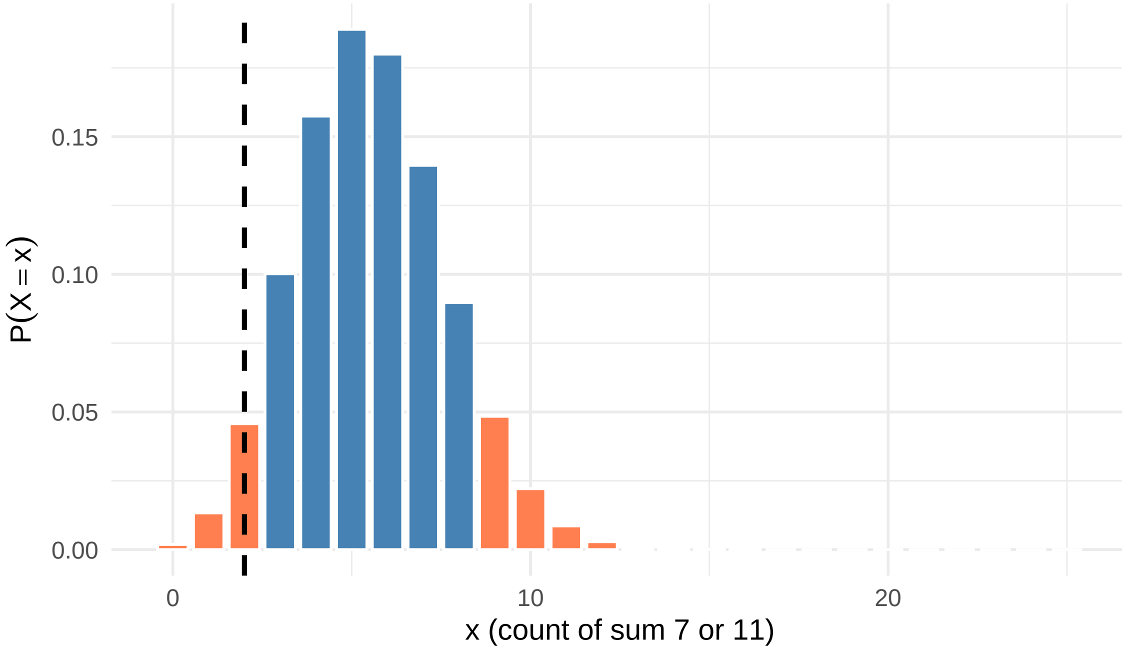


Figure 3: Example 3: dice sum 7 or 11 — two-sided p-value region shaded, x = 2

**p-value:** $P(X \leq 2) + P(X \geq 9)$ under $H_0$. For example, `pbinom(2, 25, 8/36) + (1 - pbinom(8, 25, 8/36))` $\approx 0.04$. Small p-value.

**Conclusion:** Reject $H_0$. The data are inconsistent with fair dice having probability 8/36 for sum 7 or 11; we have evidence that $p \neq 8/36$.

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## Example 4

**Research question:** Do more than 20% of students at our school use the gym at least once per week?

**Parameter:** $p$ = probability that a randomly chosen student uses the gym at least once per week (i.e., the proportion of such students in the population).

**Hypotheses:** $H_0 : p = 0.20$ vs $H_a : p > 0.20$

$n$ and $p$ under $H_0$: $n = 25$ students, $p = 0.20$

**BRP conditions (brief):** Binary (uses gym weekly or not), fixed $n = 25$, fixed $p$ under null, independent students.

**Data:** In a sample of 25 students, $x = 5$ use the gym at least once per week.

**Null plot:** $H_a : p > 0.20$ — shade where $x \geq 5$ (same side as > in the alternative). Dashed line at observed $x = 5$.

### Code anatomy — Null plot with p-value shading (Project 1 pattern)

| Part | Role |
|---|---|
| n `25` | Number of trials under $H_0$ |
| p `0.20` | Success probability under $H_0$ |
| x_obs `5` | Observed count from **your data** |
| in_pvalue | Logical vector: `TRUE` where bar is in p-value region ($x \geq$ observed (right tail)) |
| geom_col(aes(fill = in_pvalue)) | Bar heights from `dbinom`; color marks p-value region |
| geom_vline(xintercept = x_obs) | Vertical line at **observed** $x$ |

| Part | Role |
|---|---|
| Sum of coral bar heights | Approximates the p-value (check with `pbinom`) |

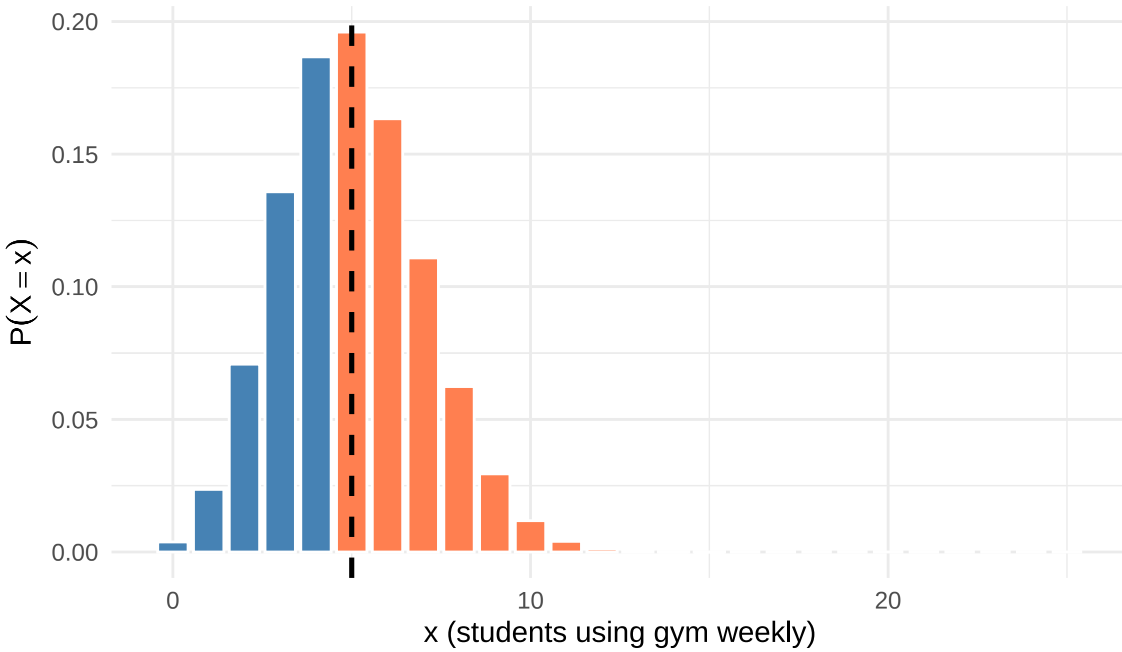


Figure 4: Example 4: gym use — p-value region shaded for Ha: p > 0.20, x = 5

**p-value:** $P(X \geq 5 \mid H_0) = 1 - P(X \leq 4)$. For example, `1 - pbinom(4, 25, 0.20)` $\approx 0.58$. Large p-value.

**Conclusion:** Fail to reject $H_0$. The data are consistent with a 20% gym-use rate; we do not have evidence that more than 20% of students use the gym at least once per week.

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## Summary

| Example | $H_a$ direction | Observed $x$ | p-value | Conclusion |
|---|---|---|---|---|
| 1 | $p > 0.10$ | 6 | Small | Reject $H_0$ |
| 2 | $p < 0.15$ | 3 | Large | Fail to reject |
| 3 | $p \neq 8/36$ (two dice, sum 7 or 11) | 2 | Small | Reject $H_0$ |
| 4 | $p > 0.20$ | 5 | Large | Fail to reject |

In every case, **steelblue** bars form the full null model, **coral** bars show the p-value region (matching the direction of $H_a$), and the **dashed line** marks observed $x$. The p-value is the sum of probabilities for the coral bars — compare that total to 1 when judging small vs large. ::: {.callout-tip icon=false} ## Check yourself — Ten-step workflow and worked examples

1. Left-handed example: $H_0: p = 0.10$ vs $H_a: p > 0.10$, observed $x = 6$. Which tail defines the p-value?
2. Defective items: $H_a: p < 0.15$, observed $x = 3$. Which tail defines the p-value?
3. Two-sided dice example: why shade **both** tails when $H_a: p \neq 8/36$?
4. What R function computes $P(X \leq k)$ for $X \sim \text{Binomial}(n, p)$?

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**Answers — Ten-step workflow and worked examples**

1. Right tail: $P(X \geq 6 \mid H_0)$.
2. Left tail: $P(X \leq 3 \mid H_0)$.
3. "More extreme than" means far from the null in **either** direction.
4. `pbinom(k, n, p)` (with default `lower.tail = TRUE`).